VARUN KUSHWAH

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TECHNICAL SKILLS

Languages: Java (Java 17/21, OOP, Multithreading, Concurrency, Stream API, Collections), SQL, TypeScript, JavaScript, Python

Backend & Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, WebSockets

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis

DevOps & Tools: Git, GitHub, Docker, Postman, Linux/Bash, Maven, Gradle, CI/CD

Core Competencies: Data Structures & Algorithms, Object-Oriented Design (OOD), Microservices Basics, Distributed Systems Basics

EDUCATION

KIET Group of Institutions 2024 – 2028
Bachelor of Technology in Computer Science & Engineering **(CGPA: 8.27 / 10.0)** *Ghaziabad, India*

New St. Stephens Public School 2021 – 2023
Senior Secondary (CBSE XII: 80%) · Secondary (CBSE X: 82%) *Agra, India*

EXPERIENCE & LEADERSHIP

devup — College Tech Club Aug 2024 – Present
Java & DSA Coordinator · Technical Mentor *Campus*

- **Spearheaded Java curriculum** and structured hands-on backend modules for 20+ active student members, accelerating peer readiness for software engineering internships and technical assessments.
- **Conducted weekly algorithmic problem-solving sessions** on trees, graphs, and dynamic programming, mentoring juniors to improve average coding speed and problem-solving accuracy by 30%.
- **Orchestrated code reviews & Git workflows**, enforcing clean architecture patterns, OOP principles, and test-driven development practices across collaborative cohort projects.

KEY PROJECTS

MentorSync — LeetCode & Codeforces LMS Live Demo | GitHub

Tech Stack: Java 21, Spring Boot 4, React 19, TypeScript, Redis 7+, MongoDB, WebSockets, Resilience4j, Docker

- **Architected a scalable two-sided LMS** for bootcamps to automate LeetCode & Codeforces tracking, assignment distribution, and real-time leaderboards.
- **Built automated verification engine** consuming LeetCode GraphQL/REST APIs and regex parsing to validate submissions, eliminating manual grading overhead.
- **Engineered nightly Redis cache warming worker** with atomic distributed locks (setIfAbsent) ensuring 0ms peak latency, STOMP WebSockets, and Resilience4j circuit breakers.

W2W Share — Offline Encrypted P2P File Transfer Platform Live Demo | GitHub

Tech Stack: Java, Spring Boot, WebRTC DataChannels, WebSockets, AES-256-GCM, Docker, Prometheus

- **Engineered an offline peer-to-peer file transfer platform** using WebRTC RTCDataChannels over UDP, bypassing central storage servers and eliminating cloud relay bandwidth costs.
- **Implemented high-concurrency Spring Boot WebSocket signaling** server to orchestrate SDP offer/answer handshakes, STUN/ICE candidate negotiation, and room connection lifecycles.
- **Engineered 64KB chunk streaming protocol** with AES-256-GCM authenticated encryption and client backpressure control, reliably streaming 10GB+ files with a steady <50MB RAM footprint.

Stateless Authentication & RBAC Service GitHub

Tech Stack: Java, Spring Boot, Spring Security, JWT, OAuth 2.0, PostgreSQL, Docker

- **Designed a modular stateless authentication service** implementing JWT access/refresh token rotation and OAuth 2.0 social sign-in, eliminating server-side session memory overhead.
- **Configured custom Spring Security filter chains** with granular Role-Based Access Control (RBAC), securing 15+ REST endpoints with zero unauthorized access vulnerabilities.
- **Integrated relational persistence layer** with connection pooling, automated validation filters, and comprehensive integration testing via Postman.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- **500+ Problems Solved on LeetCode:** Demonstrated strong algorithmic mastery across arrays, strings, trees, dynamic programming, and graph algorithms.
- **Top 105 Nationally — Google Solution Challenge 2025:** Recognized among the top 105 projects nationwide out of thousands of submissions for real-world engineering impact.